

Data Science project task 2

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# Import the data from the specified path
youth_not_eet <- read.csv("/Users/vasilikimichalia/Desktop/IDS_Group_Proj-main/Exported_data/youth_not_eet.csv")
# Rename the fourth column for clarity
colnames(youth_not_eet)[4] <- "Percentage"

# 1. Define youth_summary and continent_colors
# Calculate the weighted average data
youth_summary <- youth_not_eet %>%
  group_by(Continent, Year) %>%
  # Use na.rm = TRUE in weighted.mean to handle missing data
  summarise(weighted_average = weighted.mean(Percentage, Population, na.rm = TRUE)) %>%
  ungroup()

## `summarise()` has grouped output by 'Continent'. You can override using the
## `.groups` argument.

# Define the color mapping vector
continent_colors <- c(
  "Africa"      = "#F8766D",
  "Asia"        = "#B79F00",
  "Europe"      = "#00BA38",
  "North America" = "#00BFC4",
  "Oceania"     = "#619CFF",
  "South America" = "#F564E3"
)

# 1. Calculate all slopes, p-values, and annotation positions
slope_data <- youth_summary %>%
  # Group the data by continent so that the linear model is fitted separately for each group
  group_by(Continent) %>%
  # Use 'do' to perform a non-standard evaluation (fitting the model) on each group
  do({
    # Fit the linear model (Ordinary Least Squares) where weighted_average is the outcome
    # and Year is the predictor variable.
    lm_fit <- lm(weighted_average ~ Year, data = .)

    # Use 'tidy' from the 'broom' package to neatly extract model coefficients and statistics
    tidy_fit <- broom::tidy(lm_fit)

    # Get the row corresponding to the 'Year' term (which contains the slope and p-value)
    slope_stats <- tidy_fit %>% filter(term == "Year")

    # Extract the slope estimate (coefficient for the Year variable)
    slope <- slope_stats$estimate
    # Extract the p-value associated with the slope (to test H0: slope = 0)
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p_value <- slope_stats$p.value

# Determine the significance notation based on standard alpha levels (0.05, 0.01, 0.001)
significance_label <- case_when(
  p_value < 0.001 ~ "(p < 0.001)", # Highly significant
  p_value < 0.01 ~ "(p < 0.01)",   # Very significant
  p_value < 0.05 ~ "(p < 0.05)",   # Significant at the 5% level
  TRUE ~ "(n.s.)"                 # Not significant (p >= 0.05)
)

# Create the final annotation label string, including the formatted slope and significance
slope_label <- sprintf("Slope: %s%.3f %s",
  ifelse(slope >= 0, "+", ""), # Add a '+' sign for positive slopes
  slope,
  significance_label)

# Return a new data frame with the label and continent (no x/y needed for this method)
data.frame(
  Continent = unique(.Continent),
  slope_label = slope_label
)
}) %>%
# Ungroup the data to allow it to be used in the ggplot call later
ungroup()

# Section 2: Combined Plot Generation

# 2. Create the combined plot using facet_wrap
# Start the ggplot object, using the full summary data and mapping Year/Percentage to axes
combined_plot <- ggplot(youth_summary, aes(x = Year, y = weighted_average)) +

# Add point geom, mapping the point color to the 'Continent' variable
geom_point(aes(colour = Continent), alpha = 0.7, size = 3) +

# Add smoothing line (linear model, show standard error), mapping color/fill to 'Continent'
geom_smooth(aes(colour = Continent, fill = Continent), method = "lm", se = TRUE, size = 1) +

# Add the slope text annotations using the pre-calculated 'slope_data'
geom_text(data = slope_data,
  # Place annotation in the top-left corner inside each facet
  aes(label = slope_label, colour = Continent),
  x = -Inf, y = Inf,
  hjust = -0.1, vjust = 1.2,
  size = 3.5, fontface = "bold") +

# Apply the custom hexadecimal color codes to the 'colour' aesthetic
scale_colour_manual(values = continent_colors) +
# Apply the custom hexadecimal color codes to the 'fill' aesthetic (for confidence interval)
scale_fill_manual(values = continent_colors) +

# Split the single plot into multiple panels based on the 'Continent' variable
facet_wrap(~ Continent,
  # 'free_y' allows each panel to have its own vertical axis range for better visualization

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scales = "free_y") +

# Define the plot labels (axes and title)
labs( x = "Year",
      y = "Weighted Average Percentage",
      # Use \n for a line break in the title and define the significance notation
      title = "Youth Not in EET: Trend by Continent\n(* = significant at 5% level)") +

# Apply a minimalist theme
theme_minimal() +
# Customize theme elements
theme(
  plot.title = element_text(size = 14, hjust = 0), # Title text size and left-alignment
  legend.position = "none", # Hide the legend as colors are continent-specific
  strip.text = element_text(size = 10, face = "bold") # Format the facet panel labels (Continent name)
)

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

# Display the final combined plot object
combined_plot

## `geom_smooth()` using formula = 'y ~ x'
```

Youth Not in EET: Trend by Continent (* = significant at 5% level)

